Gauri Kshirsagar

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PROFESSIONAL SUMMARY

MSc Data Science student at the **University of Leicester** with strong experience in data analytics, data cleaning, and building scalable data workflows. Skilled in Python, SQL, Power BI, and Excel for transforming large datasets into structured insights. Experienced in developing machine learning models, dashboards, and automated data pipelines, with a growing focus on AI and LLM-based systems.

EDUCATION:

University of Leicester: MSc in Data science | Sep2025 -Present

SPPU: Bachelor of Engineering (B.E.) in Artificial Intelligence and Data Science. | Dec 2021 -Jun 2025

INTERNSHIP:

Data Analytics Intern

IBM Skill Build (CSRBOX) | 15 Jul 2024 – 5 Aug 2024

- Analysed and pre-processed structured datasets using data wrangling techniques to improve analysis accuracy.
- Implemented exploratory data analysis (EDA) and generated visual insights to identify trends and correlations.
- Presented key analytical findings to support data-driven decision-making and reporting processes.

Cybersecurity Intern

AICTE | Jul 2024 – Sep 2024

- Conducted vulnerability assessments across 5+ simulated lab environments, analysing common attack vectors including phishing, malware, and access misconfiguration.
- Prepared and documented 20+ security vulnerabilities, improving threat detection accuracy and risk visibility through structured reporting.
- Facilitated implementation of mitigation strategies aligned with cybersecurity best practices, enhancing compliance awareness and incident response readiness

PROJECTS:

Fraud Detection System | Python | R | scikit-learn | pandas | Power BI | Ongoing

- Built a scalable Python pipeline to ingest and validate 75,000+ transaction records, automating data quality checks across 28 engineered features with 0.91+ ROC-AUC accuracy
- Engineered structured features and developed dashboards to support **reliable risk analysis and communicate validated insights**

Subjective Answer Evaluation System | Python, NLP, Machine Learning

- Achieved **85% semantic similarity accuracy** across **850+ student responses**, reducing manual grading time by **40%** through NLP-based automated answer evaluation.
- Engineered TF-IDF and word embedding features and evaluated **5 machine learning algorithms** (Logistic Regression, SVM, Random Forest, KNN, Naïve Bayes), selecting the optimal model using **F1-score, accuracy, and cross-validation**.

Energy Consumption Analysis | R| Data Modelling | EDA | March 2026

- Prepared and transformed **10,000+ industrial energy records**, performing data validation to maintain accuracy, consistency, and reliability across time-series datasets
- Analysed key relationships (e.g., **r ≈ 0.89**) and consumption patterns to support **data-driven insights and energy efficiency improvements (5–10%)**.

Socio-Demographic Dashboard (OAC 2021) | R, Tableau | Feb2026

- Authored 2021 **Census OAC** dataset covering **8 Supergroups** and **100+ Output Areas**, integrating **4 key demographic variables** (age structure, education levels, ethnic diversity, household composition) into a coordinated multi-view dashboard.
- Designed **4 interactive visualisations** (100% stacked bar chart, grouped bar chart, boxplot, scatterplot) with district-average benchmarking and reference lines to improve comparative policy-level interpretation.

CERTIFICATIONS:

- International Conference Paper Presentation – ICIA-MET'25 (April 2025)
- Data Science, Python, ML, DL & NLP Bootcamp – Udemy (99 Hours, March 2025)

TECHNICAL SKILLS:

- **Programming:** Python (pandas, NumPy, scikit-learn, data transformation, automation, pipeline development), R (tidyverse, ggplot2)
- **Databases:** SQL, MySQL
- **Machine Learning And NLP:** Classification, Regression, Model Evaluation, Feature Engineering, Exposure to LLMs and NLP-based systems (text similarity, embeddings)
- **Data Analytics And BI:** EDA, Statistical Modelling, Advanced Excel (Pivot Tables, Power Query, Data Cleaning)
  Power BI (Data Modelling, DAX, Dashboard Development, Data Refresh), Tableau
- **Analytical Focus:** Predictive modelling, trend analysis, optimisation thinking, decision support